BMCS2074 · ARTIFICIAL INTELLIGENCE

CONSUMER COMPLAINT URGENCY CLASSIFICATION

Financial complaint triage with classical NLP and DistilBERT

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01

PROBLEM

COMPLAINT TEAMS MUST DISTINGUISH INCONVENIENCE FROM CONTINUING HARM

Routine issue

> “The transfer was briefly held, then released after review.”

A frustrating event, but no continuing material impact.

Time-sensitive harm

> “Unauthorized transactions are still occurring and essential funds are inaccessible.”

The narrative signals ongoing financial and account-access risk.

Product category and sentiment are insufficient. The model must interpret the full narrative.

URGENCY IS DEFINED BY IMMEDIACY AND SEVERITY

Low

Routine or resolved
Informational questions, minor inconvenience, or dissatisfaction without continuing material impact.

Medium

Significant and unresolved
Priority attention is needed, but the narrative does not show explicit immediate severe harm.

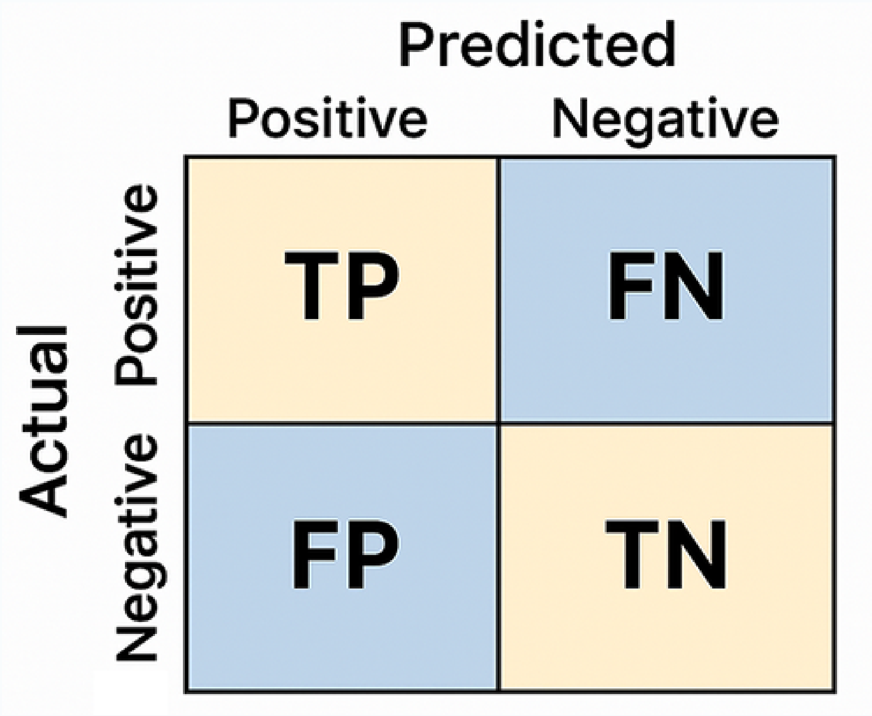
High

Immediate or continuing risk
Financial, identity, housing, legal, safety, or essential account-access harm is active or imminent.

WE EVALUATED MORE THAN HEADLINE ACCURACY

01

How effectively can four NLP methods classify Low,Medium, and High urgency?



02

How do they compare on macro-F1, High recall, and serious error directions?

03

What performance, interpretability, confidence,and cost trade-offs matter?

THE MISSING PIECE WAS A CONTROLLED, HARM-BASED URGENCY STUDY

PRIOR EMPHASIS

- Product classification
- Company response prediction
- Topic discovery
- General complaint detection

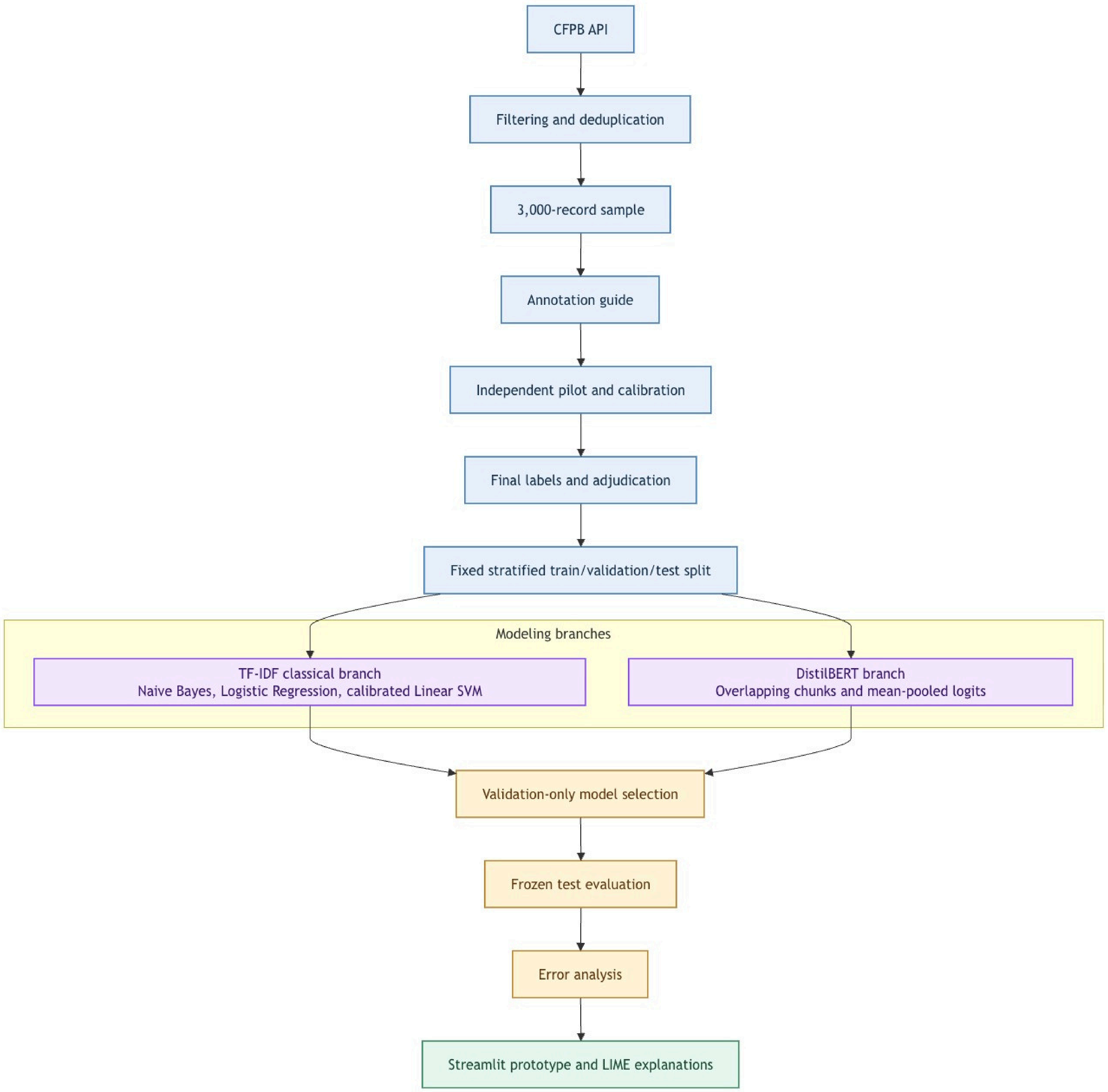
OUR CONTRIBUTION

Three-level urgency based on current harm, evaluated with identical splits and connected to a human-led operations prototype.

02

METHODOLOGY

OUR PIPELINE



WE PRODUCED 3,000 UNIQUE, LABELED NARRATIVES

6,000
Raw complaints retrieved through the CFPB API

−1,445
Duplicate narratives discarded

4,555
Unique narratives in the cleaned pool

3,000
Seeded final sample for annotation and modeling

2022–2025
Complaint receipt window

SHA-256
Processed hashes recorded

20260806
Final sampling seed

REFINING THE GUIDE INCREASED CHANCE-CORRECTED AGREEMENT

INITIAL PILOT · 400 RECORDS

κ **0.4278**

83.25% raw agreement

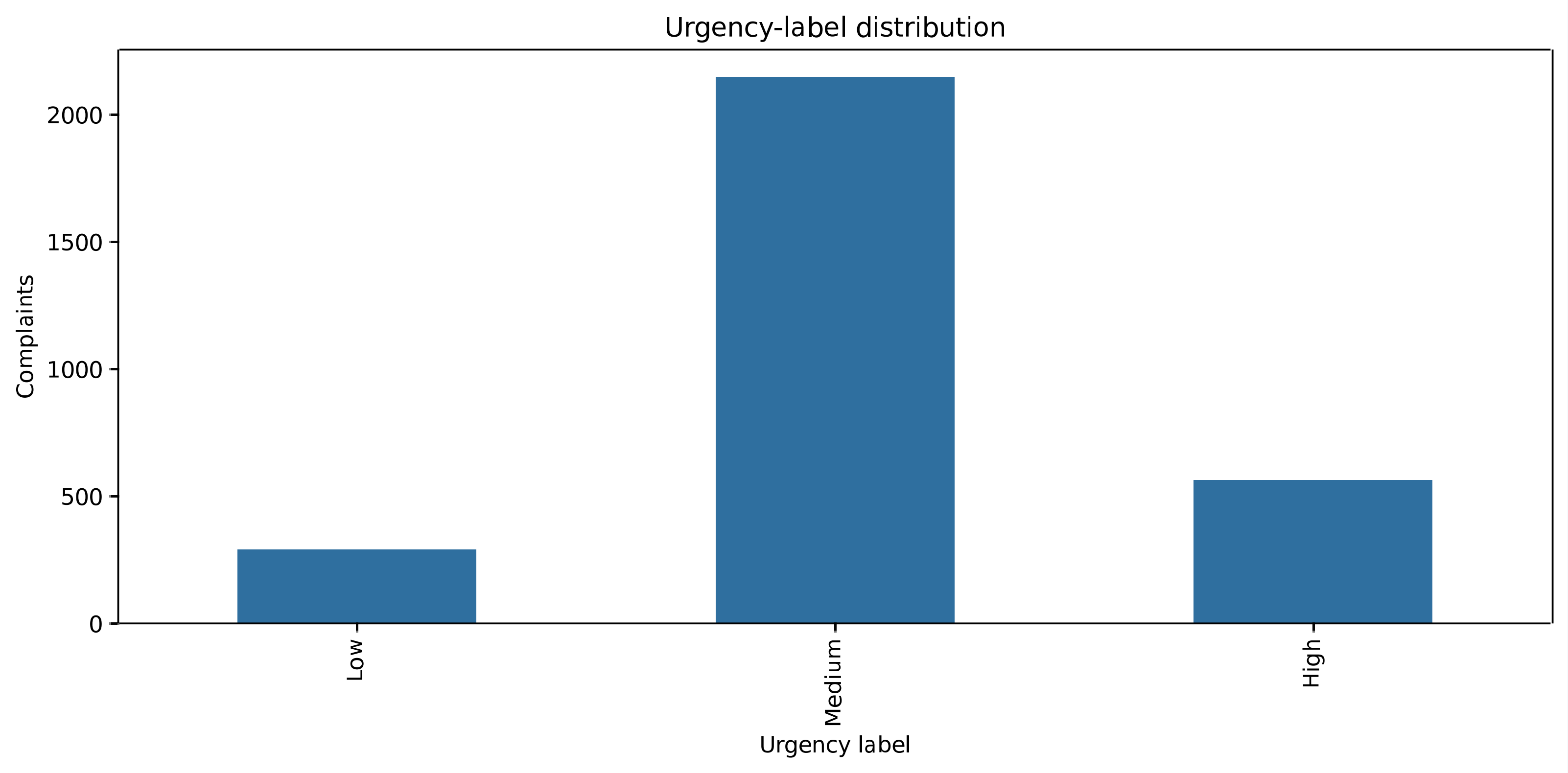
CALIBRATION · 67 RECORDS

κ **0.7485**

83.58% raw agreement

> Two independent AI annotators (gpt5.6-terra_low) produced research labels; six uncertain final cases were reviewed and every decision retained provenance.

CLASS IMBALANCE MADE ACCURACY AN UNSAFE PRIMARY METRIC

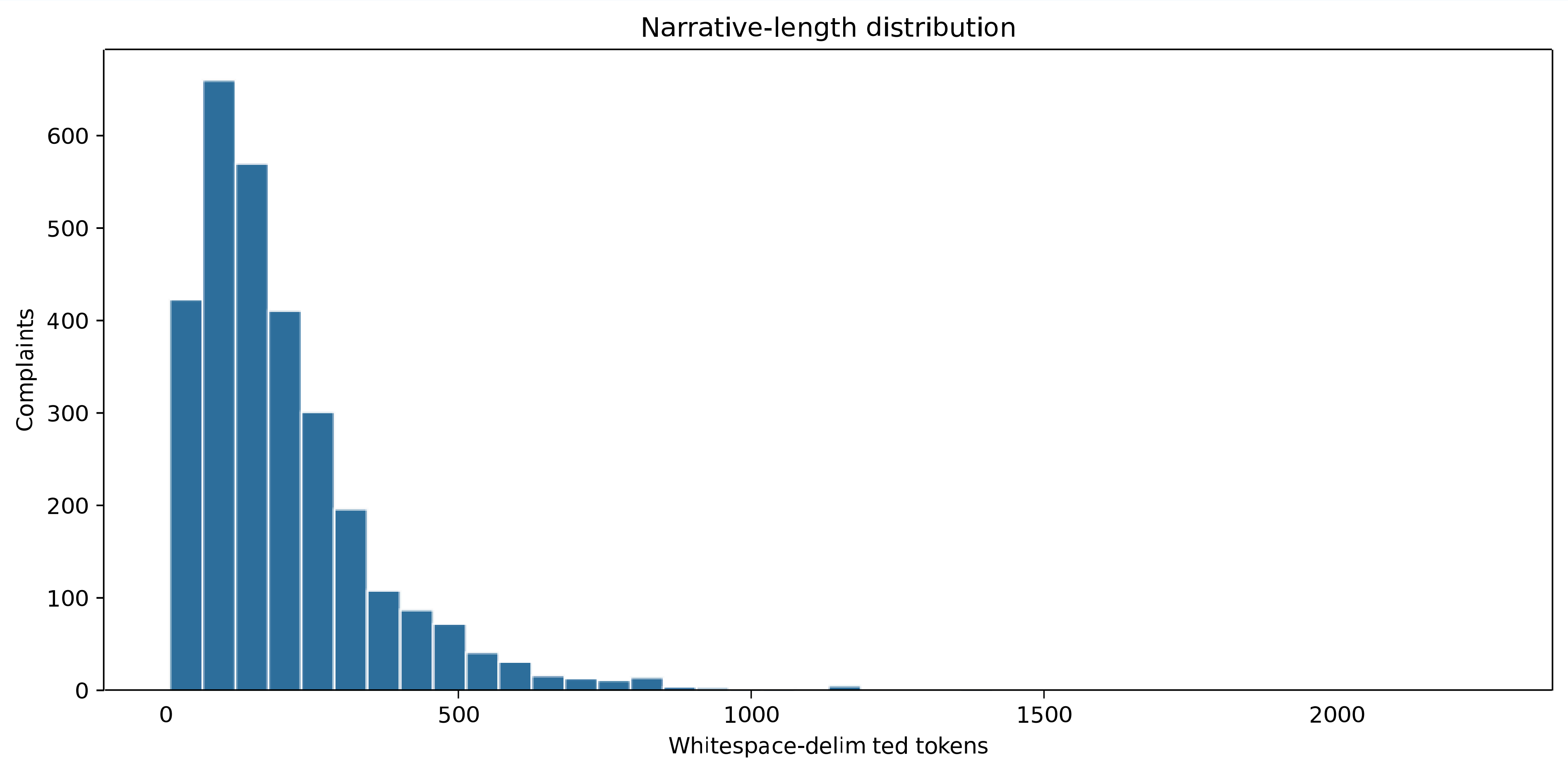


n = 3,000 labeled narratives.

Low	9.67%
Medium	71.60%
High	18.73%

> Macro-F1 gives each urgency class equal influence.

LONG COMPLAINTS REQUIRED AN EXPLICIT DOCUMENT STRATEGY FOR DISTILLBERT



43.1%

of training narratives exceeded 256 tokens

DistilBERT used overlapping 256-token chunks with a stride of 64, then averaged chunk logits.

CLASSICAL AND CONTEXTUAL MODELS SAW THE SAME NARRATIVES DIFFERENTLY

CLASSICAL PATH

27,493 TF-IDF features

- Lowercase and placeholder normalization
- Unigrams and bigrams
- Training-fitted vocabulary
- Full normalized narrative

TRANSFORMER PATH

Contextual token sequences

- DistilBERT uncased tokenizer
- Minimal text alteration
- Overlapping long-document chunks
- Mean-logit document aggregation

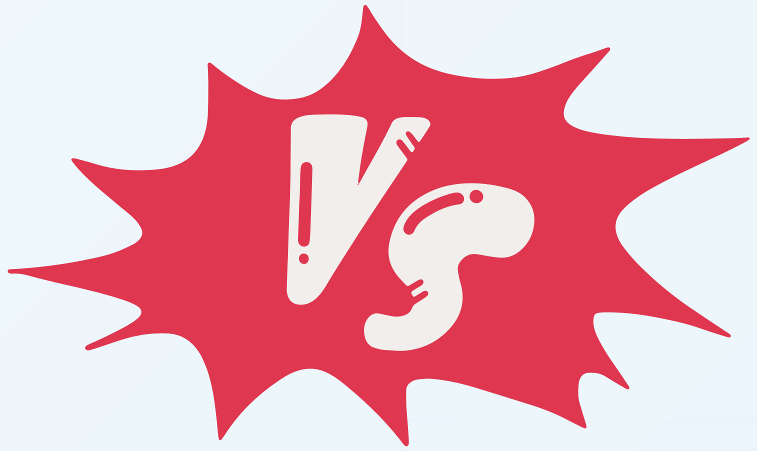
FOUR MODELS TESTED DISTINCT ASSUMPTIONS ABOUT LANGUAGE

Multinomial Naive Bayes

Fastprobabilisticbaselinefornon-negative sparse features.

Logistic Regression

Balanced,regularized,probabilistic linear baseline.



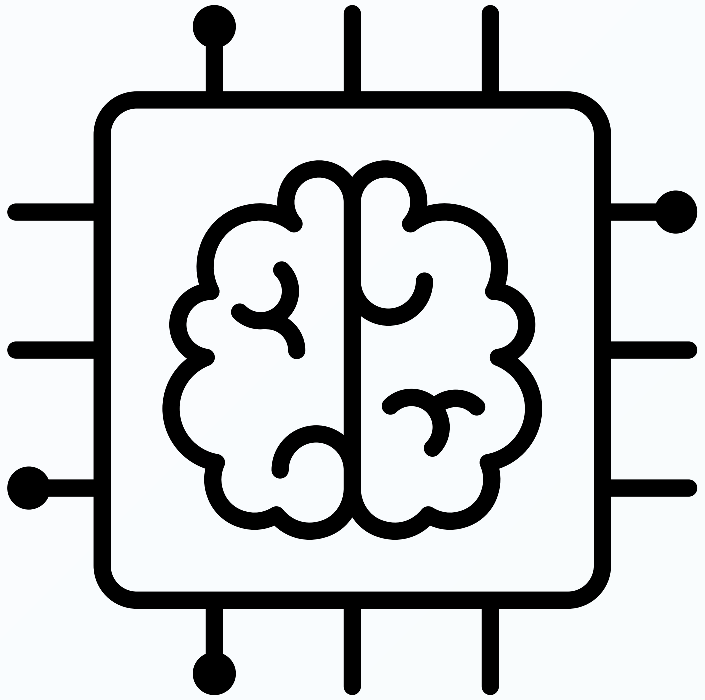
Calibrated Linear SVM

Margin-basedtextclassifierwith five-fold sigmoid calibration.

DistilBERT

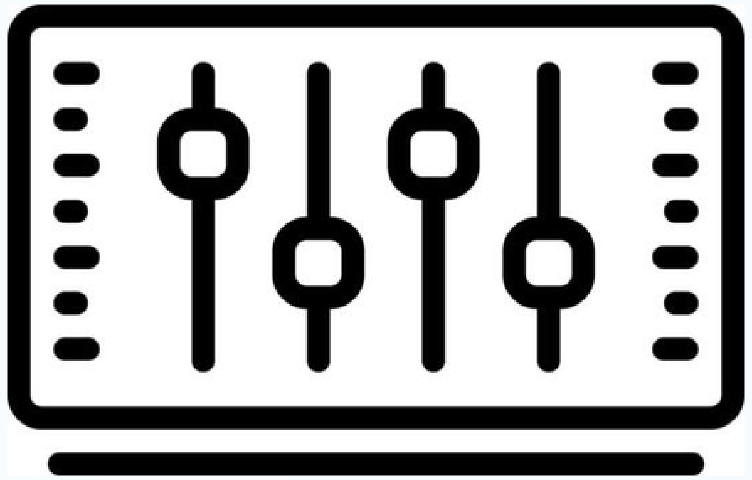
Fine-tunedtransformer testing whether context improves class balance.

OUR SPLITTING CONFIG



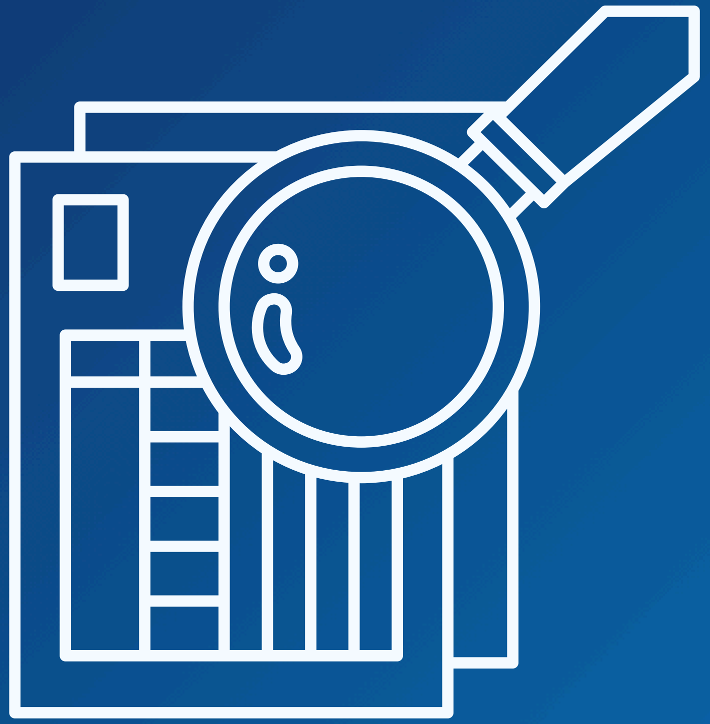
70%

Fit features, weights, parameters, and calibration folds



15%

Select configurations by macro-F1



15%

450 held-out test
Opened once after freezing

CONFIGURATIONS WERE SELECTED ON VALIDATION MACRO-F1

MODEL	SELECTED CONFIGURATION
Naive Bayes	alpha = 0.1
Logistic Regression	C = 1.0 · balanced class weights · max_iter = 5000
Calibrated Linear SVM	C = 1.0 · balanced weights · five-fold sigmoid calibration
DistilBERT	LR 2e-5 · batch 8 · five epochs · 256 tokens · stride 64 · inverse-frequency weights

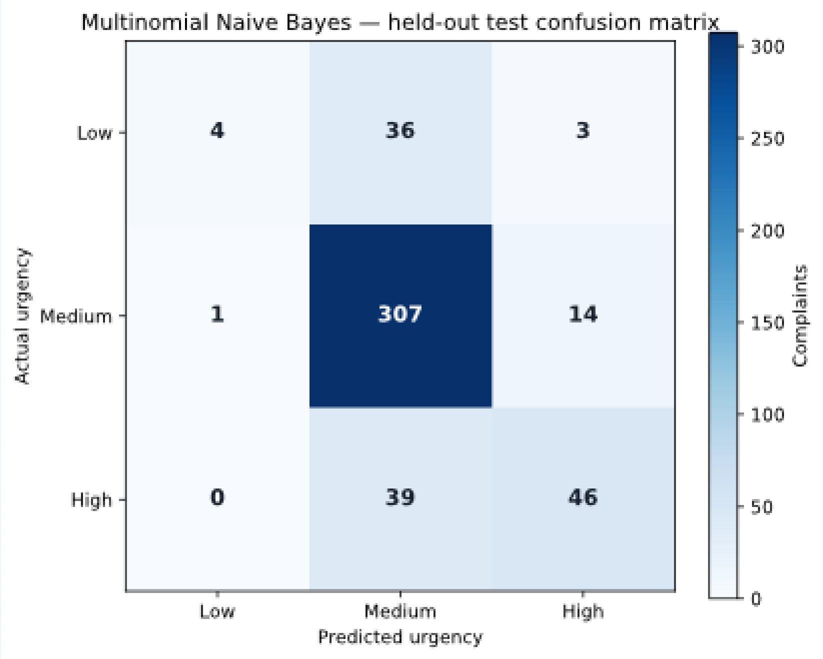
03
RESULTS & INTERPRETATION

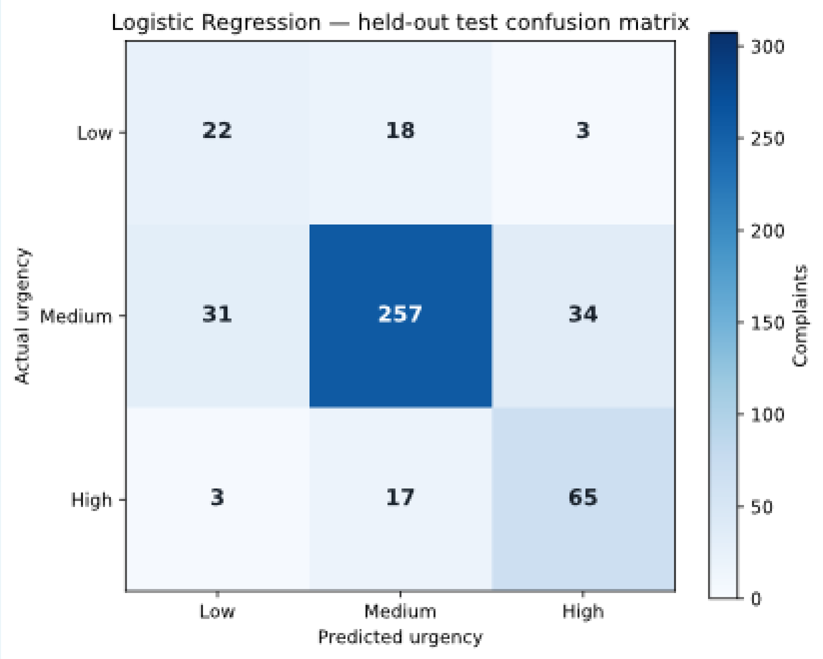
FULL HELD-OUT TEST COMPARISON

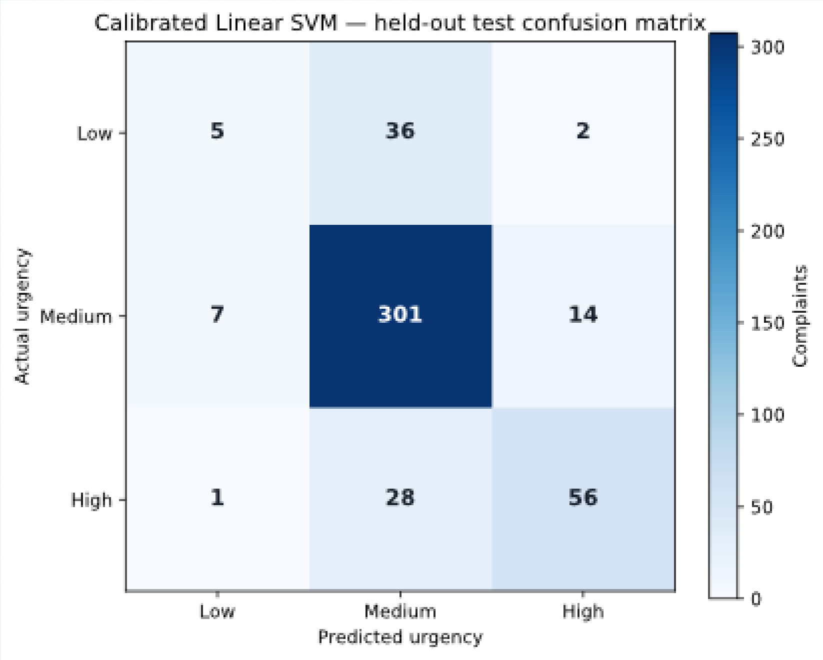
MODEL	ACCURACY	MACRO-F1	WEIGHTED-F1	HIGH RECALL	HIGH→LOW
Naive Bayes	0.7933	0.5535	0.7574	0.5412	0
Logistic Regression	0.7644	0.6589	0.7728	**0.7647**	3
Calibrated Linear SVM	**0.8044**	0.5894	0.7 78 8	0.6588	1
DistilBERT	0.7 73 3	**0.6 747**	**0.7807**	0.7059	2

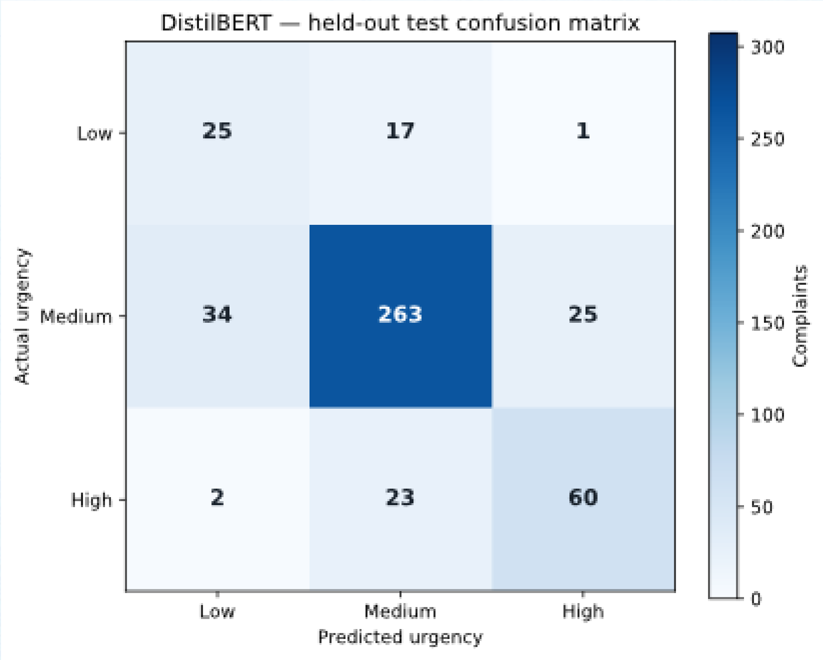
Held-out test set: $n = 450$.

CONFUSION MATRICES

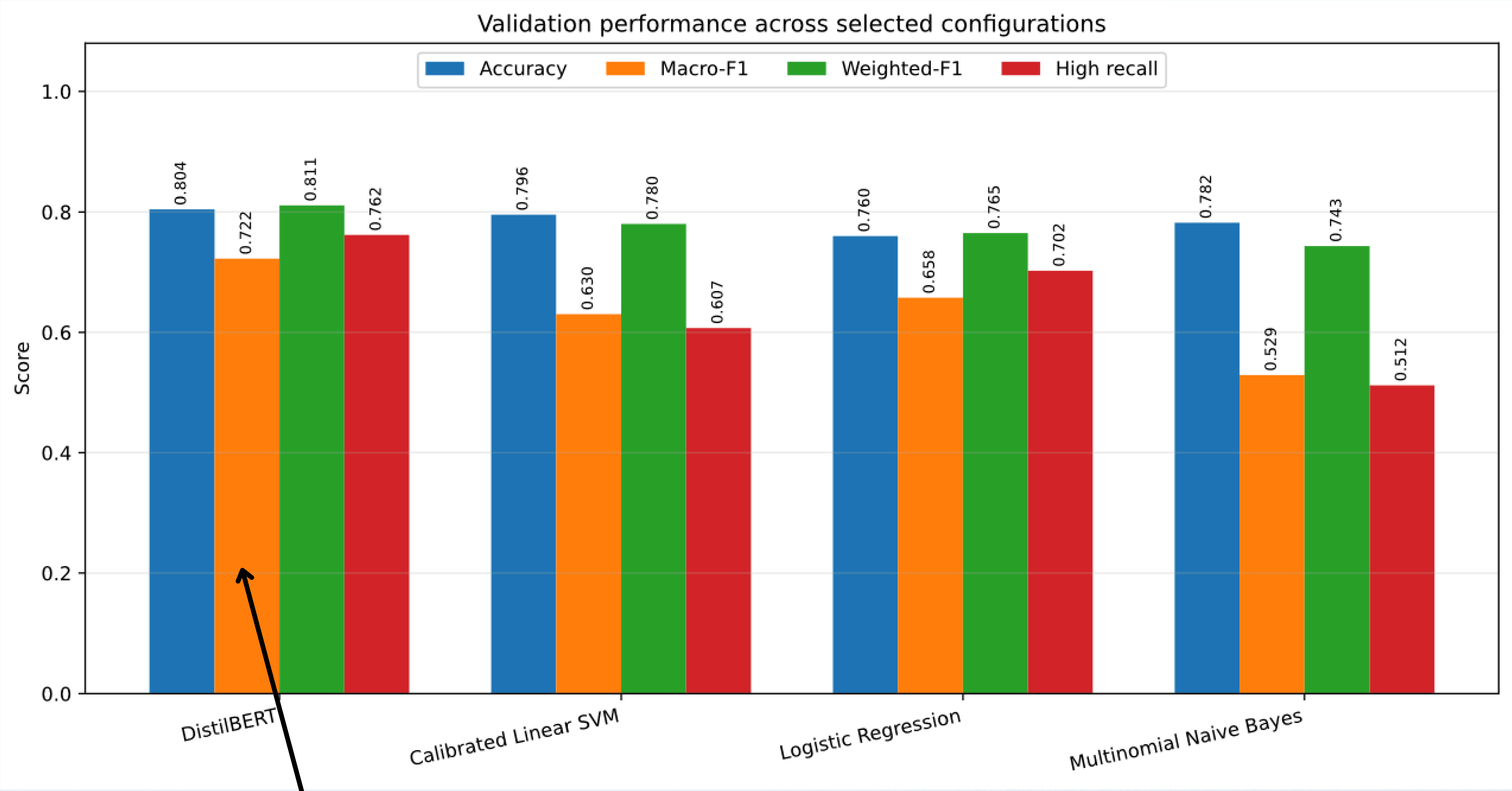






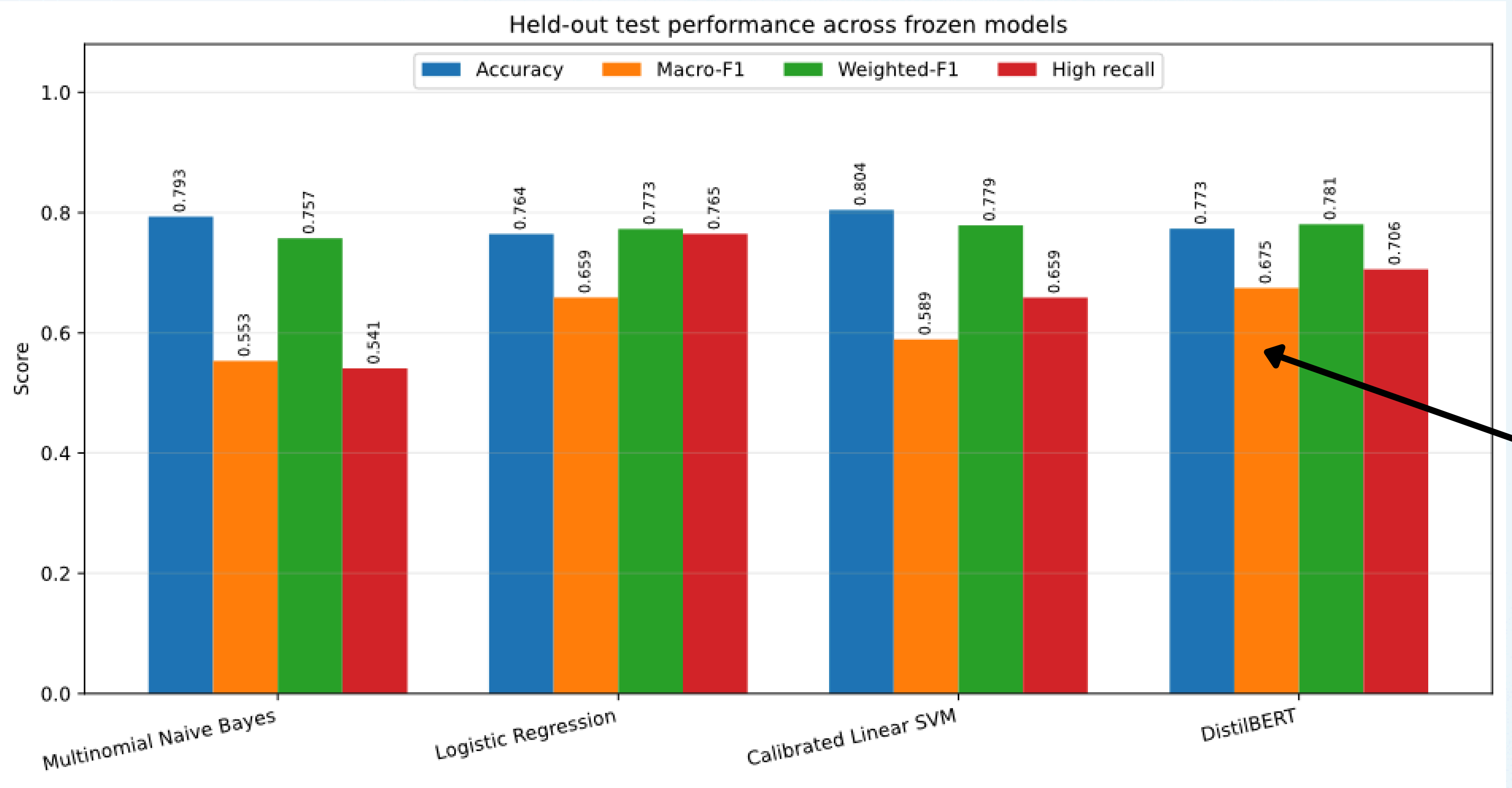


VALIDATION FAVORED DISTILBERT'S CLASS BALANCE



Validation macro-F1: **DistilBERT 0.7225** · Logistic Regression 0.6576 · SVM 0.6302 · Naive Bayes 0.5290

DISTILBERT ACHIEVED THE BEST BALANCED HELD-OUT PERFORMANCE



PRIMARY METRIC

0.6747

DistilBERT macro-F1
+0.0158 over Logistic Regression

Same 450 complaint IDs for every frozen model; no post-test tuning.

BUT THE HIGHEST-ACCURACY MODEL WAS NOT THE BEST MODEL

CALIBRATED SVM

0.8044

Highest test accuracy

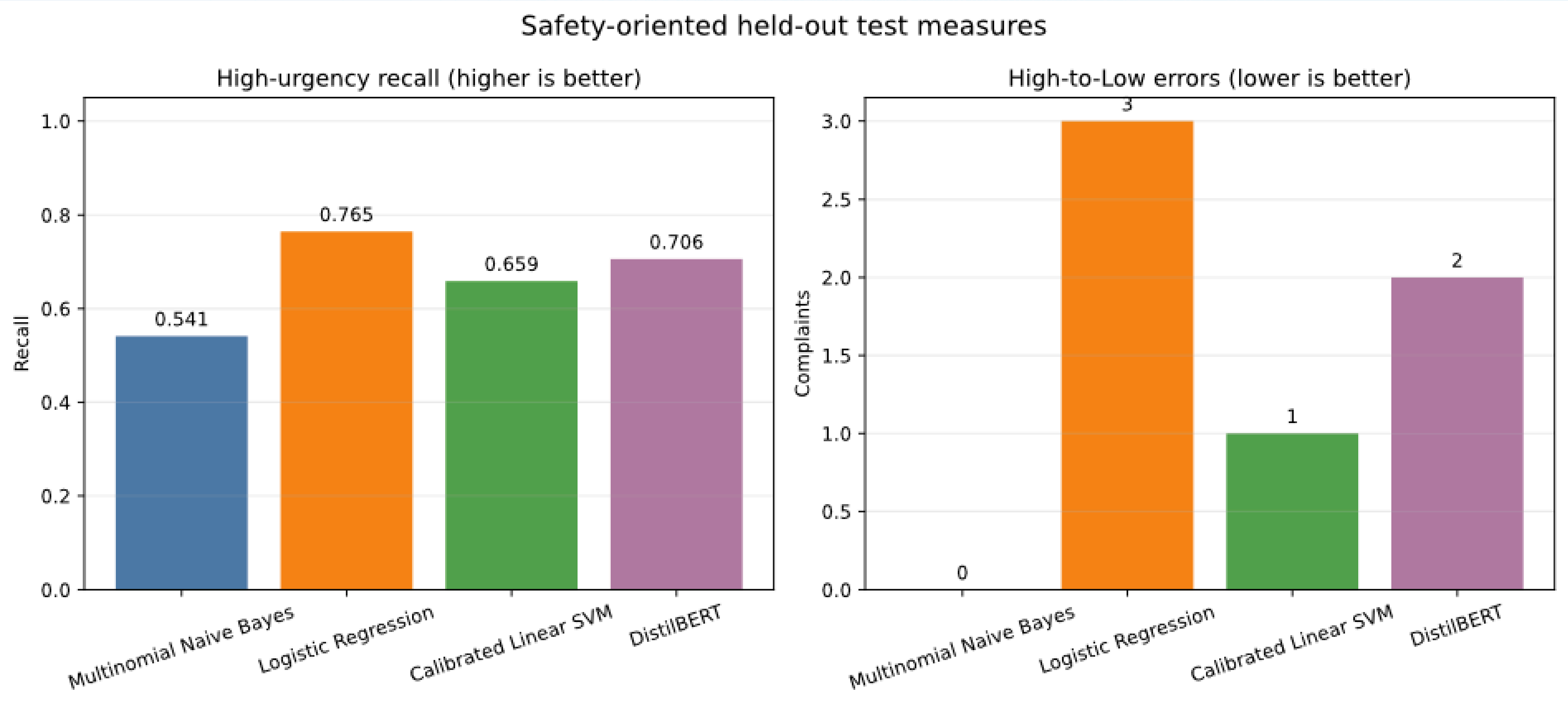
BUT FOR LOW URGENCY

0.1786

Low-class F1·only 5 of 43 Low cases recovered

...Majority-class success can conceal operationally important failure.

SAFETY METRICS CHANGED HOW THE MODELS WERE INTERPRETED

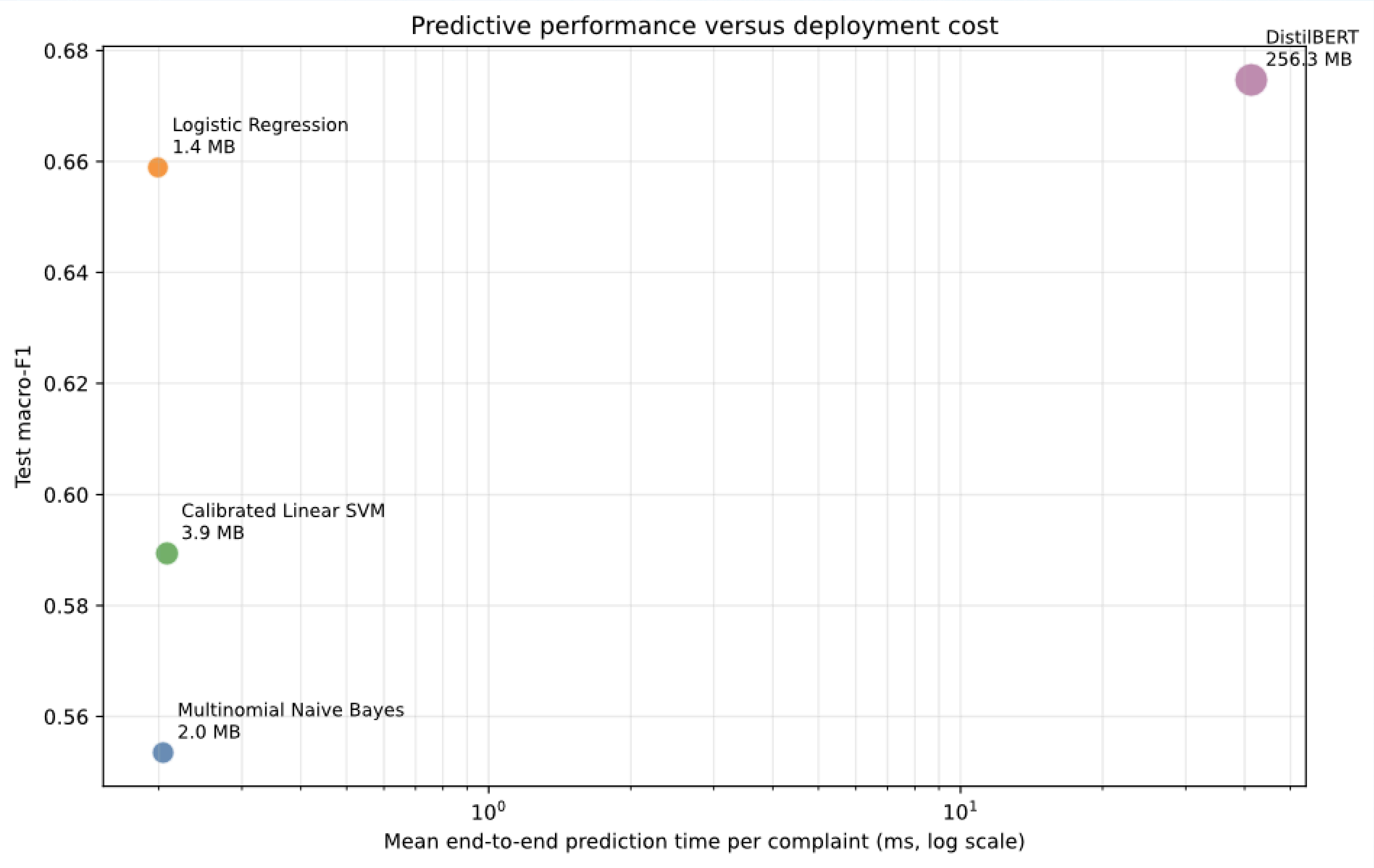


0.7647

Best High recall
Logistic Regression

Zero direct High→Low errors did not make Naive Bayes safe: 39 High cases were predicted Medium.

THE PERFORMANCE GAIN CARRIED A SUBSTANTIAL COMPUTATIONAL COST



207×

slower mean prediction than Logistic Regression

189×

larger persisted artifact set

ERRORS CLUSTERED AROUND IMPLICIT RISK, TIMING, AND LONG CONTEXT

Actual High → predicted Low

Implicit identity misuse

An unauthorized inquiry was described as potential identity misuse, but routine credit-report language outweighed the urgency cue.

Actual High → predicted Medium

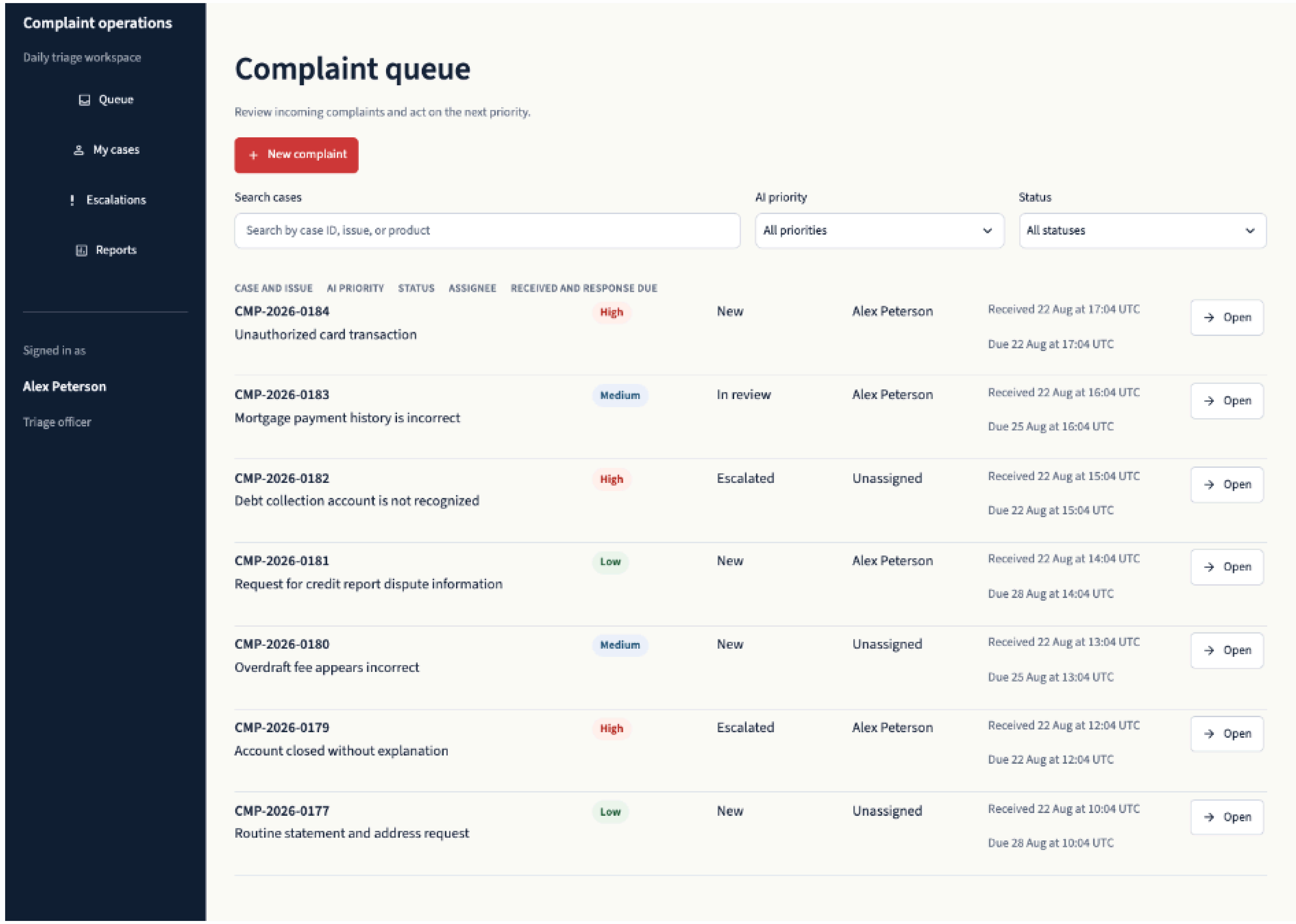
Urgency diluted by background

A long mortgage narrative mixed history, allegations, and continuing risk, making the decisive passage harder to isolate.

Borderline and lengthy complaints remain strong candidates for mandatory human review.

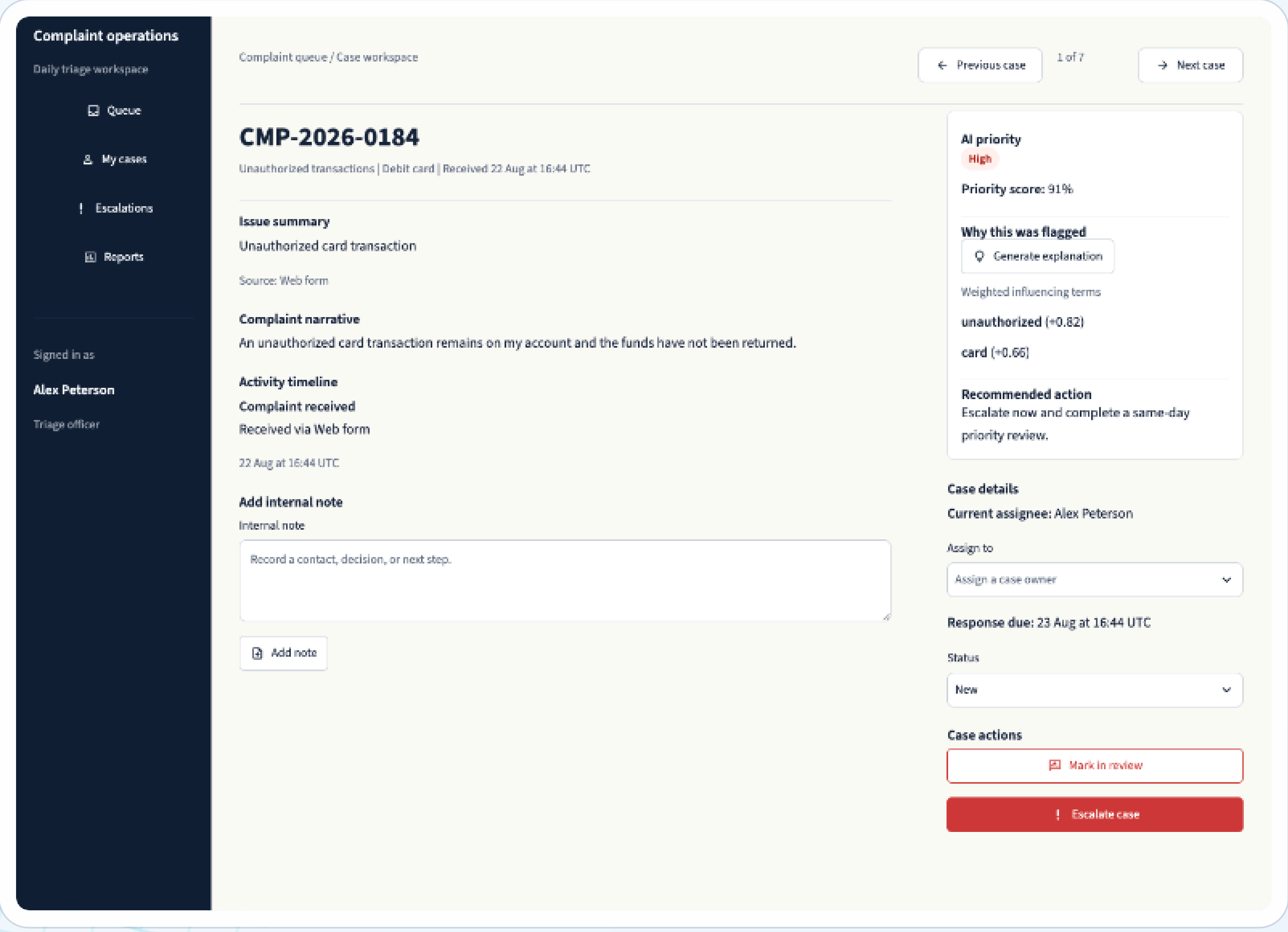
04
PROTOTYPE SHOWCASE

THE PROTOTYPE BEGINS WITH AN OPERATIONAL COMPLAINT QUEUE



All visible case data are synthetic.

PREDICTIONS BECOME INPUTS TO A HUMAN-LED CASE WORKFLOW



- AI priority and score
- Recommended action
- Assignee and status controls
- Internal notes and timeline
- On-demand “Why this was flagged” support

> The officer remains responsible for the operational decision.

05

DECISION

THE RIGHT RECOMMENDATION DEPENDS ON THE OPERATING PRIORITY

PERFORMANCE RECOMMENDATION

DistilBERT

- Best macro-F1: 0.6747
- Strongest Low-class F1
- Selected for the academic prototype

LIGHTWEIGHT ALTERNATIVE

Logistic Regression

- Macro-F1: 0.6589
- Best High recall: 0.7647
- 1.36 MiB and ~207× faster prediction

DistilBERT’s advantage is real but modest—not decisive under every deployment constraint.

STRONGER LABELS AND UNCERTAINTY HANDLING ARE THE NEXT PRIORITIES

CURRENT BOUNDARIES

- LLM-generated urgency labels
- One held-out test of 450 cases
- Only 43 Low and 85 High test examples
- Non-representative, unverified
- narratives Unmeasured calibration and fairness

FUTURE WORK

- Domain-expert annotation
- Confidence intervals and external validation
- Calibration, abstention, and thresholding
- Fairness and temporal-drift analysis
- Alternative long-document aggregation

06

CONCLUSION

Context matters

Urgency cannot be reduced to product or tone

DistillBERT leads

Best balanced held-out performance.

Humans decide

AI supportstriage;itdoesnotdetermine outcomes.

THANK YOU